



SOC Internship – Week 3 Report

Deploying pfSense Firewall, Squid Proxy, SquidClamAV
and Wazuh SIEM for Malware Detection and Monitoring

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This report documents the design, deployment, and validation of a small-scale Security Operations Center (SOC) monitoring environment built entirely in a virtual lab, combining perimeter firewalling, web-proxy content inspection, antivirus scanning, and SIEM-based log correlation.

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Introduction

Modern organizations depend on layered, defense-in-depth network security to protect their systems and data from an ever-growing range of cyber threats. No single control is sufficient on its own: firewalls restrict and filter network traffic, web proxies inspect and control outbound browsing activity, antivirus engines scan content for known malicious signatures, and Security Information and Event Management (SIEM) platforms tie all of these data sources together so that security analysts can detect, investigate, and respond to incidents from a single pane of glass.

This report documents a hands-on lab project completed during Week 3 of the SOC internship at ITSimplera Solutions. The objective was to build a small but realistic security stack inside a virtualized environment and to prove, end to end, that a malicious download could be detected, blocked, logged, forwarded, correlated, and alerted on – exactly as it would be in a production SOC workflow. The stack consists of four components working together:

- **pfSense Firewall** – acts as the network gateway, routes traffic between the internal lab network and the simulated internet (WAN), and generates firewall logs for every connection it evaluates.
- **Squid Proxy** – sits between client machines and the internet, intercepting and inspecting all outbound HTTP/HTTPS traffic before it leaves the network.
- **SquidClamAV** – integrates the ClamAV antivirus engine with Squid so that every file downloaded through the proxy is scanned in real time and blocked if it matches a known malware signature.
- **Wazuh SIEM** – centrally collects logs from pfSense over Syslog, applies detection rules (including a custom rule written specifically for this project), and surfaces the resulting alerts on an analyst-facing dashboard.

To validate the pipeline, network traffic was generated from a Parrot Security OS client, malicious content was simulated using the industry-standard EICAR test file, and the resulting logs and alerts were traced all the way from the firewall to the Wazuh dashboard. The remainder of this report walks through the objectives, the lab architecture, the step-by-step configuration, the testing performed, and the results and lessons learned.

1 Project Objectives

The primary objectives of this project were:

- Deploy and configure a pfSense firewall in a virtual environment to act as the network gateway and traffic-filtering point.
- Configure Squid Proxy so that all client web traffic is routed through it for inspection before reaching the internet.
- Integrate SquidClamAV with Squid to automatically detect and block malicious file downloads in real time.
- Configure Remote Syslog on pfSense to forward firewall logs to the Wazuh server for centralized analysis.
- Deploy and configure Wazuh SIEM to collect, parse, and index logs from across the environment.

- Develop and validate a custom Wazuh detection rule specifically for pfSense firewall events, so that matching traffic automatically generates an alert.
- Verify the complete detection workflow – from traffic generation to dashboard alert – using the safe, industry-standard EICAR malware test file.

2 Project Workflow

The diagram below summarizes the end-to-end sequence followed to build and validate the lab, from initial virtual machine setup through to reviewing alerts on the Wazuh dashboard. Each stage builds on the one before it: the network and firewall had to exist before the proxy could route through it, the proxy had to be in place before antivirus scanning could be added, and logging had to be working before the custom detection rule could be tested.

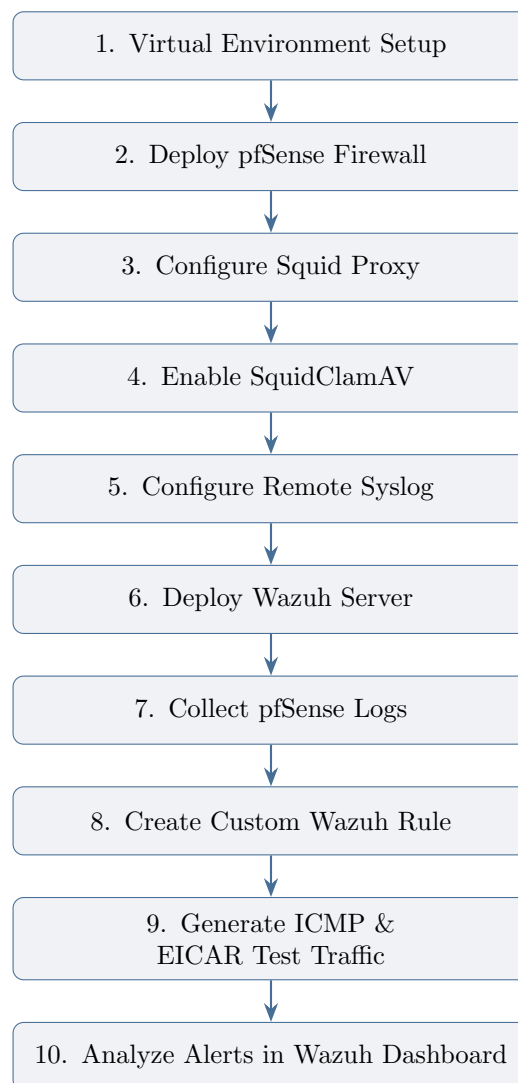


Figure 1: End-to-end project workflow, from lab setup to alert analysis.

3 Lab Environment

The table below lists the core software and platforms used to build the lab, and the figure that follows shows the running virtual machines inside VirtualBox – pfSense acting as the gateway,

the Wazuh server providing the SIEM back end, and the Parrot Security OS client used to generate test traffic.

Component	Configuration
Firewall	pfSense 2.8.1
Proxy	Squid Proxy
Antivirus	SquidClamAV
SIEM	Wazuh 4.14.6
Client	Parrot Security OS
Host	Windows 11
Virtualization	Oracle VirtualBox

Table 1: Core lab environment components.

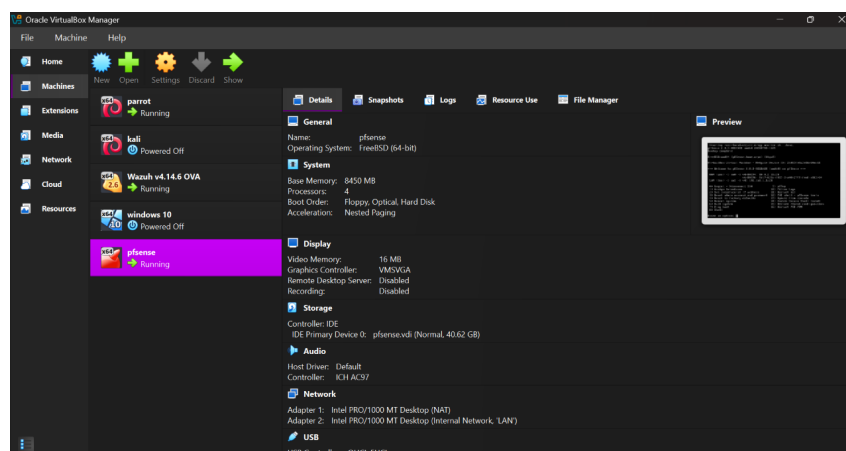


Figure 2: VirtualBox Manager showing the virtual machines that make up the lab: the pfSense gateway, the Wazuh SIEM server, and the Parrot Security OS client, all connected on an internal VirtualBox network.

4 Network Topology

The diagram below shows the network topology used for this lab. The pfSense firewall sits at the center of the design, acting as the single point through which all traffic must pass between the internal LAN (where the Parrot OS client and Wazuh server live) and the external WAN (representing the internet). Placing the firewall in this position means that every packet – legitimate or malicious – can be logged, filtered, and, where necessary, blocked before it reaches its destination.

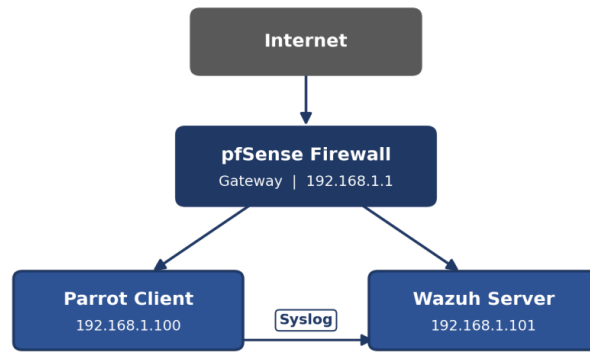


Figure 1. Network Topology - pfSense, Parrot Client, and Wazuh Server

Figure 3: Network topology: pfSense as the central gateway between the internal LAN (Parrot client, Wazuh server) and the external WAN.

4.1 System Architecture

(See Figure above for the full network topology diagram.)

Components

Component	Purpose
pfSense Firewall	Provides network firewalling, routing between LAN and WAN, and traffic filtering. Acts as the default gateway for all lab hosts.
Squid Proxy	Sits between clients and the internet, inspecting and controlling outbound web traffic before it leaves the network.
SquidClamAV	Extends Squid with real-time antivirus scanning, blocking malicious files at download time.
Wazuh Manager	Centrally collects, parses, and analyzes logs from across the environment, and generates alerts when detection rules match.
Parrot OS	The client machine used to generate legitimate and simulated-malicious network traffic for testing.
Internet (WAN)	The external network used to test connectivity and to source the EICAR malware test file for detection testing.

Table 2: Role of each component within the overall architecture.

5 Tools and Technologies

Tool	Purpose
Oracle VirtualBox	Virtualization platform used to host all lab virtual machines.
pfSense CE	Open-source firewall and network security platform.
Squid Proxy	Web proxy service used to inspect and control outbound traffic.
SquidClamAV	Antivirus integration that plugs the ClamAV engine into Squid.
Wazuh 4.14.6	Security Information and Event Management (SIEM) platform used for log collection, correlation, and alerting.
Parrot Security OS	Linux-based client operating system used to generate test traffic.
Syslog	Standard protocol used to forward pfSense firewall logs to the Wazuh server.
EICAR Test File	Industry-standard, harmless file used to safely simulate malware detection.
ICMP (Ping)	Used to generate simple, easily identifiable network traffic for testing the firewall and the custom Wazuh rule.

Table 3: Tools and technologies used throughout the project.

6 Environment Setup

This section details the exact software versions and network configuration used to stand up the lab, so that the environment can be reproduced if needed.

Component	Details
Firewall	pfSense CE 2.8.1
Proxy	Squid Proxy
Antivirus	SquidClamAV 7.3_2
SIEM	Wazuh 4.14.6
Operating System (SIEM Server)	Rocky Linux
Client Machine	Parrot Security OS
Network	VirtualBox Internal Network
Syslog	pfSense → Wazuh
Malware Test	EICAR Test File

Table 4: Detailed environment and version information.

6.1 Virtual Lab Architecture

All virtual machines were connected on a single VirtualBox internal network so that traffic between them stayed isolated from the host machine and from the outside world, giving a safe space to test malware detection without any risk of infection spreading beyond the lab.

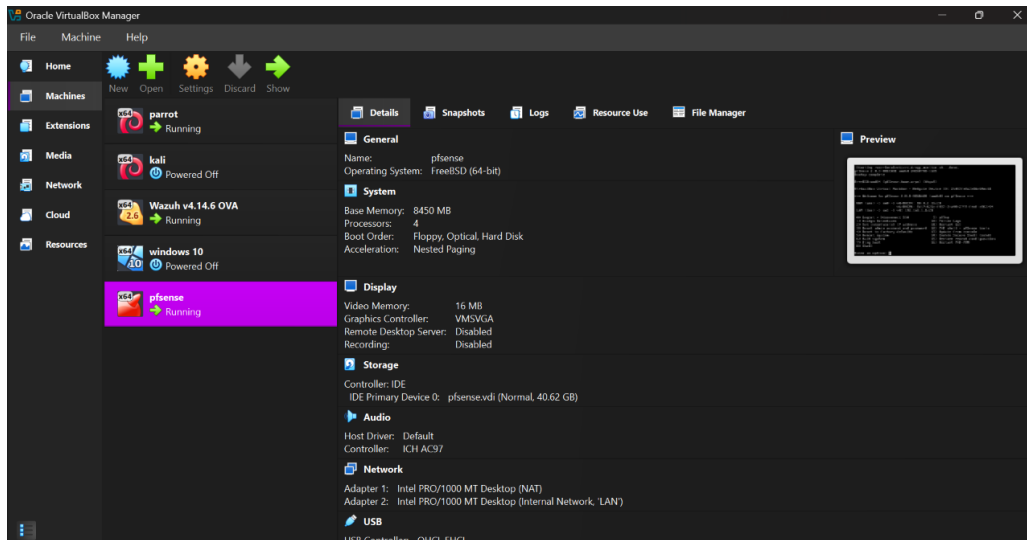


Figure 4: Figure 2. VirtualBox environment used for the project, showing the pfSense, Wazuh, and Parrot OS virtual machines running together on the internal lab network.

6.2 IP Addressing

Device	IP Address	Purpose
pfSense LAN	192.168.1.1	Gateway & Firewall
Parrot Client	192.168.1.100	Test machine
Wazuh Server	192.168.1.101	SIEM
Internet	WAN	Malware download testing

Table 5: IP address plan used across the lab.

6.3 Software Installed

Software	Purpose
pfSense	Firewall
Squid	HTTP Proxy
SquidClamAV	Malware Scanning
Wazuh Manager	Log Collection
Wazuh Dashboard	Monitoring
Wazuh Agent	Endpoint Monitoring

Table 6: Software components installed across the lab virtual machines.

6.4 Screenshots

The following screenshots show each core system running and reachable inside the lab, confirming that the environment was fully operational before testing began.

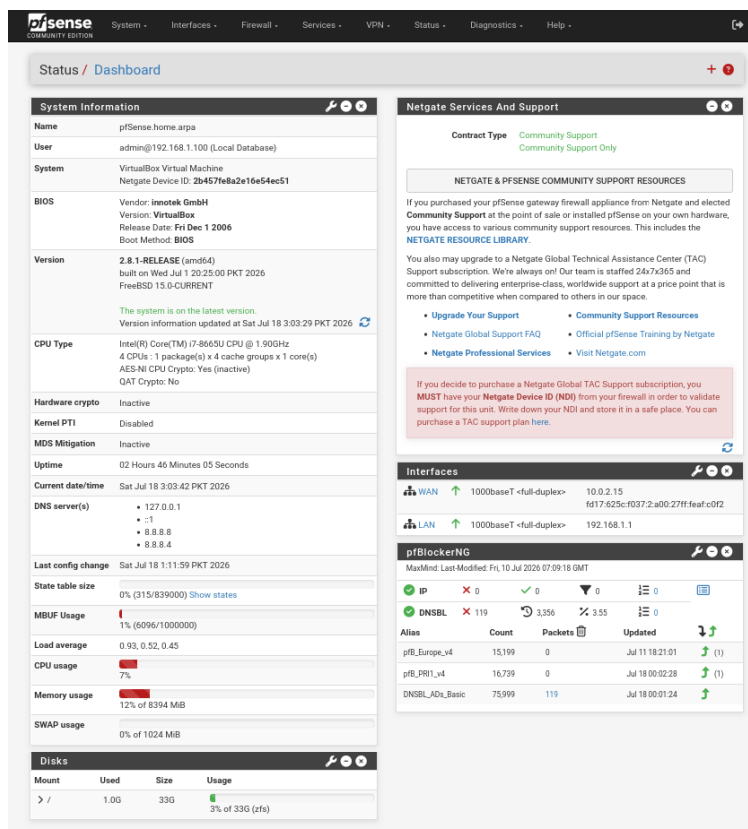


Figure 5: Figure 3. pfSense web administration dashboard, showing system status, interface information, and gateway health for the firewall.

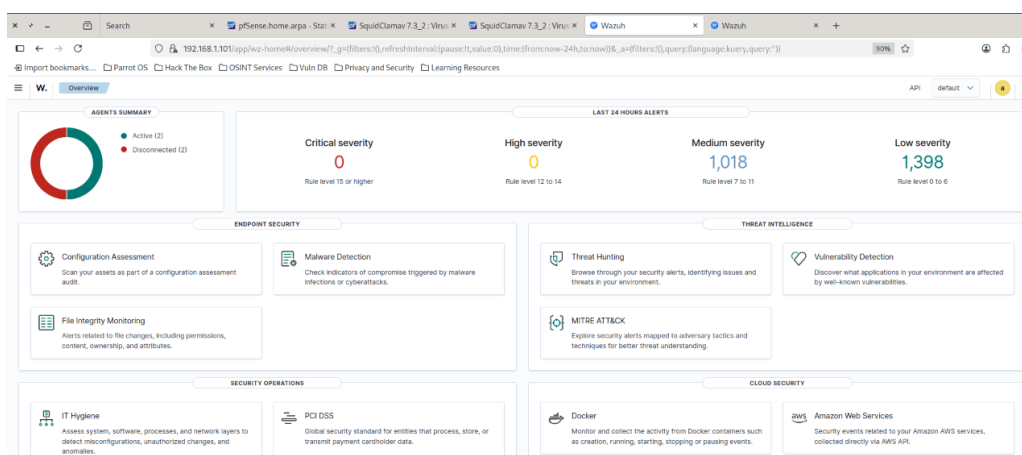


Figure 6: Figure 4. Wazuh Dashboard home view, showing the SIEM server online and ready to receive and index events from connected sources.

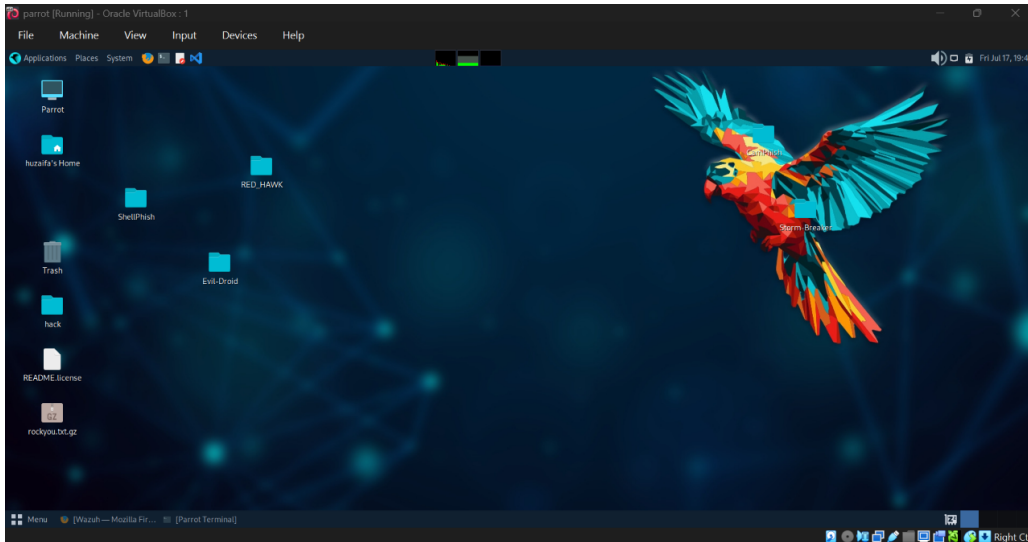


Figure 7: Figure 6. Parrot Security OS desktop, the client machine used to generate ICMP and web traffic for testing.

7 Testing and Validation

This section demonstrates that each security control was tested successfully, moving through the pipeline in the same order that traffic actually flows: firewall filtering, DNS-based web filtering, malware scanning, log forwarding, custom rule matching, and finally dashboard alerting.

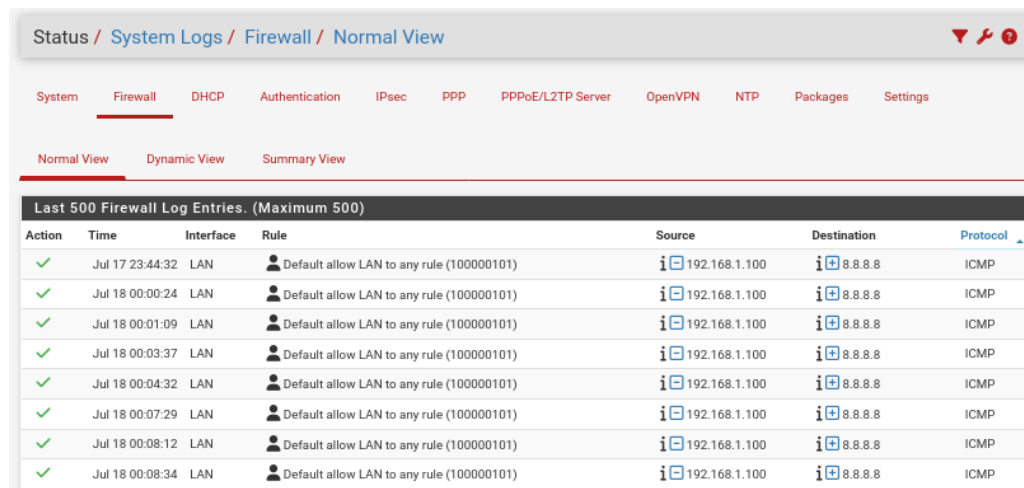
7.1 Firewall Filtering Test (pfSense)

Objective

Verify that pfSense correctly filters and logs network traffic passing through the gateway.

Procedure

- Generated ICMP (ping) traffic from the Parrot OS client.
- Observed the traffic arriving in the pfSense Firewall Logs.
- Confirmed that packets were processed according to the configured firewall rules.



The screenshot shows the pfSense Firewall Log interface. The breadcrumb navigation is 'Status / System Logs / Firewall / Normal View'. Below the navigation, there are tabs for 'System', 'Firewall', 'DHCP', 'Authentication', 'IPsec', 'PPP', 'PPPoE/L2TP Server', 'OpenVPN', 'NTP', 'Packages', and 'Settings'. Under the 'Firewall' tab, there are sub-tabs for 'Normal View', 'Dynamic View', and 'Summary View'. The main content area is titled 'Last 500 Firewall Log Entries. (Maximum 500)'. It displays a table with the following columns: Action, Time, Interface, Rule, Source, Destination, and Protocol. The table shows eight log entries, all with a green checkmark in the 'Action' column, indicating that the traffic was allowed. The 'Rule' column for all entries is 'Default allow LAN to any rule (100000101)'. The 'Source' column shows the IP address '192.168.1.100' and the 'Destination' column shows '8.8.8.8'. The 'Protocol' column shows 'ICMP' for all entries. The 'Time' column shows various timestamps from July 17 to July 18.

Action	Time	Interface	Rule	Source	Destination	Protocol
✓	Jul 17 23:44:32	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP
✓	Jul 18 00:00:24	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP
✓	Jul 18 00:01:09	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP
✓	Jul 18 00:03:37	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP
✓	Jul 18 00:04:32	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP
✓	Jul 18 00:07:29	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP
✓	Jul 18 00:08:12	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP
✓	Jul 18 00:08:34	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP

Figure 8: Figure 6.1.1. pfSense Firewall Log showing ICMP traffic being evaluated and passed by the firewall rule set.

Result: Traffic successfully passed through the firewall and was logged, confirming that pfSense was correctly positioned as the gateway and actively inspecting traffic.

7.2 DNSBL Website Blocking Test

Objective

Verify that connections to known-malicious domains are blocked by pfBlockerNG's DNS Block List (DNSBL) feature.

Procedure

- Attempted to open a domain on the DNSBL blocklist from the Parrot OS client.
- pfBlockerNG intercepted the DNS lookup and redirected the request to a local blocking page instead of resolving the real address.



Figure 9: Figure 6.1.2. DNSBL block page returned to the client, confirming the request to the blocked domain was intercepted before it could reach the internet.

Result: DNSBL successfully prevented access to the blocked domain, demonstrating that the firewall provides an additional layer of protection at the DNS level, ahead of any file-based scanning.

7.3 Malware Detection using EICAR

Objective

Validate antivirus detection using the EICAR test file, a file specifically designed by the European Institute for Computer Antivirus Research so that antivirus tools can be safely tested without using real malware.

Procedure

- Downloaded the EICAR test file from https://secure.eicar.org/eicar_com2.zip through the Parrot OS client.
- SquidClamAV intercepted and scanned the download as it passed through the Squid proxy.
- The download was blocked immediately, before the file reached the client.



Figure 10: Figure 6.3.1. SquidClamAV block page confirming the EICAR test signature was detected and the download was refused.

Result: The EICAR test signature was successfully detected and blocked, confirming that SquidClamAV was correctly integrated with Squid and actively scanning downloaded content in real time.

7.4 Log Forwarding Verification

Objective

Verify that pfSense firewall logs are correctly forwarded to, and received by, the Wazuh server.

Procedure

- Configured pfSense Remote Logging to send Syslog output to the Wazuh server's IP address.
- Enabled Syslog collection on the Wazuh manager.
- Generated firewall traffic and checked that corresponding events appeared on the Wazuh side.

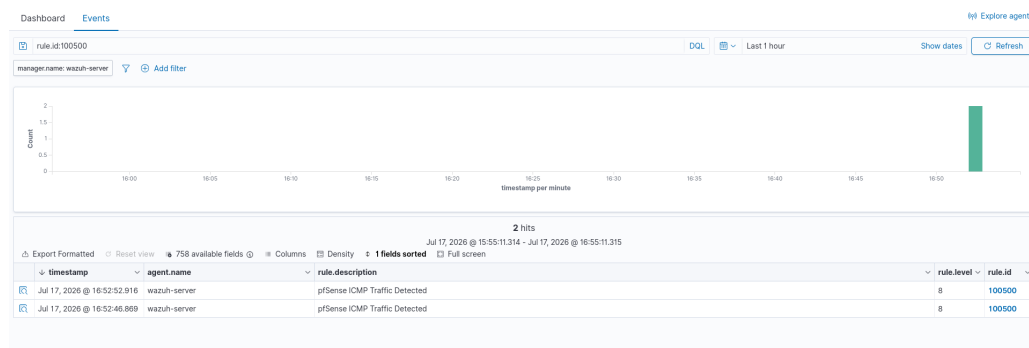


Figure 11: Figure 6.4.1. Wazuh Dashboard showing pfSense firewall events successfully received over Syslog.

Table	JSON
@timestamp	Jul 17, 2026 @ 16:52:52.916
_index	wazuh-alerts-4.x-2026.07.17
agent.id	000
agent.name	wazuh-server
data.action	pass
data.dstip	8.8.8.8
data.dstport	4653
data.id	100000101
data.length	164

W.	Discover	wazuh-alerts-4.x-2026.0717#PXnacZ8BwH9jp5MTjxaT	a
data.srcport			
decoder.name	pf		
full_log	Jul 18 01:52:51 pfsense.home.arpa filterlog[87346]: 89,,100000101,em1,match,pass,in,4,0x0,,64,61287,0,DF,1,icmp,84,192.168.1.100,8.8.8.8,reqe		
id	1784321572.926456		
input.type	log		
location	/var/log/pfsense.log		
manager.name	wazuh-server		
predecoder.hostname	pfsense.home.arpa		
predecoder.program_name	filterlog		
predecoder.timestamp	Jul 18 01:52:51		
rule.description	pfSense ICMP Traffic Detected		
rule.firedtimes	2		
rule.groups	pfsense, local		
rule.id	100500		
rule.level	8		
rule.mail	false		
timestamp	Jul 17, 2026 @ 16:52:52.916		

Figure 12: Figure 6.4.2. Expanded event detail view in Wazuh, showing the individual fields parsed out of a forwarded pfSense log entry, including source/destination IP and the firewall action taken.

Result: Firewall logs were successfully received by Wazuh, confirming that the Syslog pipeline between pfSense and the SIEM was working end to end.

7.5 Custom Rule Validation

Objective

Verify that a custom-written Wazuh rule correctly detects ICMP firewall traffic coming from pfSense.

Procedure

- Created custom rule **100500** in `local_rules.xml`, built on top of Wazuh's built-in pfSense (pf) decoder.
- Restarted the Wazuh Manager to load the new rule.
- Tested the rule offline using the `wazuh-logtest` utility before relying on live traffic.

```

wazuh-user@wazuh-server:~$ sudo /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.14.6
Type one log per line

Jul 18 01:52:51 pfsense.home.arpa filterlog[87346]: 89,,,100000101,em1,

**Phase 1: Completed pre-decoding.
  full event: 'Jul 18 01:52:51 pfsense.home.arpa filterlog[87346]
164'
  timestamp: 'Jul 18 01:52:51'
  hostname: 'pfsense.home.arpa'
  program_name: 'filterlog'

**Phase 2: Completed decoding.
  name: 'pf'
  action: 'pass'
  dstip: '8.8.8.8'
  dstport: '4653'
  id: '100000101'
  length: '164'
  protocol: 'icmp'
  srcip: '192.168.1.100'
  srcport: ''

**Phase 3: Completed filtering (rules).
  id: '100500'
  level: '8'
  description: 'pfSense ICMP Traffic Detected'
  groups: '['pfsense', 'local']'
  firetimes: '1'
  mail: 'False'

**Alert to be generated.

```

Figure 13: Figure 6.5.1. wazuh-logtest output validating that a sample pfSense log line correctly matches custom rule 100500.

```

** Alert 1784327336.1016654: - pfsense,local,
2026 Jul 17 22:28:56 pfsense.home.arpa->/var/log/pfsense.log
Rule: 100500 (level 8) -> 'pfSense ICMP Traffic Detected'
Src IP: 192.168.1.100
Src Port:
Dst IP: 8.8.8.8
Dst Port: 5450
Jul 18 03:28:56 pfsense.home.arpa filterlog[87346]: 89,,,10000
length: 164

```

Figure 14: Figure 6.5.2. Wazuh alert output confirming that the custom rule matched a live pfSense ICMP log entry and generated an alert.

Result: The custom rule successfully matched pfSense ICMP logs and generated an alert, confirming that the detection logic was both syntactically valid and functionally correct.

7.6 Wazuh Dashboard Alert Verification

Objective

Verify that the custom rule generates a visible, analyst-facing alert in the Wazuh Dashboard – the final step of the detection pipeline.

Procedure

- Generated ICMP traffic from the Parrot OS client.
- Opened the Wazuh Dashboard.
- Verified that a new alert had been created for the matching traffic.

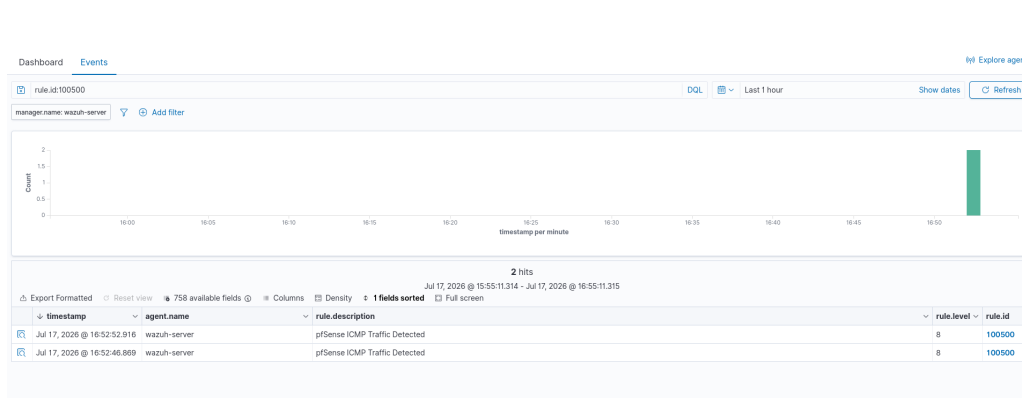


Figure 15: Figure 6.6.1. Alert entry in the Wazuh Dashboard generated by custom rule 100500.

Figure 6.6.1 shows the alert appearing successfully in the dashboard with:

- Rule ID: **100500**
- Level: **8**
- Source IP
- Destination IP
- Protocol: ICMP

7.7 Overall Testing Summary

Test	Status
pfSense Firewall	Passed
DNSBL Filtering	Passed
SquidClamAV Malware Detection	Passed
Syslog Forwarding	Passed
Wazuh Log Collection	Passed
Custom Wazuh Rule	Passed
Dashboard Alert Generation	Passed

Table 7: Summary of all tests performed and their outcomes.

8 Incident Response Plan

To demonstrate not just detection but a realistic analyst workflow, the traffic generated during testing was walked through a standard six-phase incident response process (Detection, Alert Generation, Analysis, Containment, Eradication, Recovery), plus a closing Lessons Learned review, using pfSense, Squid Proxy, SquidClamAV, and Wazuh SIEM.

Phase	Action Performed
1. Detection	Malware traffic was detected when the user attempted to download the EICAR test file through the Squid proxy. SquidClamAV scanned the file and immediately identified it as Eicar-Test-Signature , preventing the download. Simultaneously, pfSense generated firewall logs, and Wazuh collected the corresponding events.
2. Alert Generation	Wazuh processed the incoming pfSense logs using the built-in pf decoder and the custom rule (Rule ID: 100500) created during this project. The generated alert contained the source IP, destination IP, protocol (ICMP), firewall action, and event severity, allowing quick identification of the activity.
3. Analysis	The security analyst reviewed the alert in the Wazuh Dashboard, verified the source host (192.168.1.100), confirmed that the firewall action was recorded correctly, and correlated the alert with the SquidClamAV warning page that displayed the detected malware signature.
4. Containment	The malicious file download was automatically blocked by SquidClamAV before it reached the client. Since the threat was stopped at the proxy, no additional firewall changes or endpoint isolation were required during testing.
5. Eradication	The detected file was not allowed to enter the network. Log files and generated alerts were reviewed to ensure that only the test malware (EICAR) triggered the detection and that no additional suspicious activity was observed.
6. Recovery	Network connectivity and security services (pfSense, Squid Proxy, SquidClamAV, Syslog, and Wazuh Manager) were verified to be functioning normally after testing. The environment returned to its normal operational state without affecting legitimate traffic.
7. Lessons Learned	The implementation successfully demonstrated end-to-end malware monitoring. pfSense generated security logs, SquidClamAV blocked malicious content, Syslog forwarded events to Wazuh, and Wazuh produced searchable alerts in the dashboard. The custom detection rule was validated successfully and confirmed that the monitoring pipeline worked as expected.

Table 8: Incident response walkthrough for the simulated EICAR detection event.

9 Results & Validation

This section summarizes the successful implementation and testing of the security solution as a whole, followed by a small set of key screenshots that together tell the full detection story: a malicious file is blocked, firewall traffic is logged, the custom rule matches, and the alert appears on the dashboard.

Component	Validation Performed	Status
pfSense Firewall	Firewall traffic monitored successfully	Passed
Squid Proxy	Client web traffic routed through proxy	Passed
SquidClamAV	EICAR test file detected and blocked	Passed
Syslog	pfSense logs forwarded to Wazuh	Passed
Wazuh SIEM	Logs collected and indexed successfully	Passed
Custom Rule (100500)	ICMP firewall traffic detected	Passed
Wazuh Dashboard	Alert generated and displayed	Passed

Table 9: Final validation results across every component of the pipeline.

Evidence: the four screenshots below form a short visual chain of custody for the test, from the moment the malicious file was blocked through to the alert reaching the analyst dashboard.



Figure 16: EICAR test file detected and blocked by SquidClamAV at the proxy.

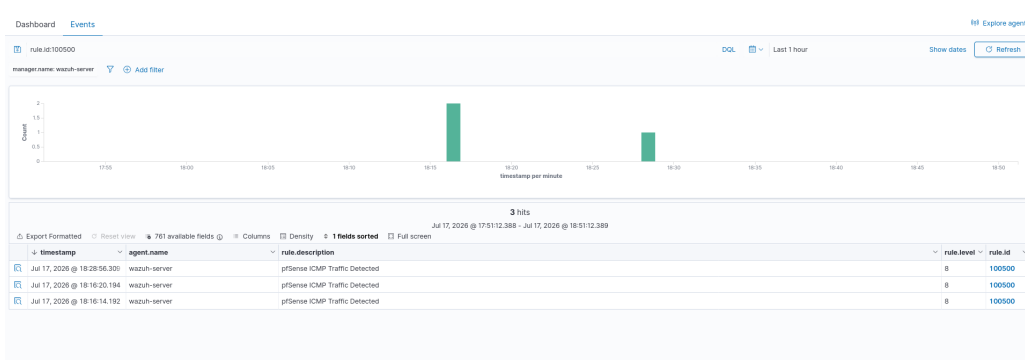


Figure 17: pfSense firewall traffic monitored successfully, confirming the gateway was actively logging connections during the test.

```

** Alert 1784327336.1016654: - pfsense,local,
2026 Jul 17 22:28:56 pfsense.home.arpa->/var/log/pfsense.log
Rule: 100500 (level 8) -> 'pfSense ICMP Traffic Detected'
Src IP: 192.168.1.100
Src Port:
Dst IP: 8.8.8.8
Dst Port: 5450
Jul 18 03:28:56 pfsense.home.arpa filterlog[87346]: 89,,,10000
length: 164

```

Figure 18: Wazuh custom rule 100500 matched the pfSense ICMP log entry, generating an alert.

Status / System Logs / Firewall / Normal View							
System Firewall DHCP Authentication IPsec PPP PPPoE/L2TP Server OpenVPN NTP Packages Settings							
Normal View Dynamic View Summary View							
Last 500 Firewall Log Entries. (Maximum 500)							
Action	Time	Interface	Rule	Source	Destination	Protocol	
✓	Jul 17 23:44:32	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	
✓	Jul 18 00:00:24	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	
✓	Jul 18 00:01:09	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	
✓	Jul 18 00:03:37	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	
✓	Jul 18 00:04:32	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	
✓	Jul 18 00:07:29	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	
✓	Jul 18 00:08:12	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	
✓	Jul 18 00:08:34	LAN	Default allow LAN to any rule (100000101)	192.168.1.100	8.8.8.8	ICMP	

Figure 19: ICMP firewall traffic detected and recorded in the pfSense firewall log.

10 Challenges Faced

Challenge	Solution
Syslog events were not appearing in Wazuh.	Verified the Remote Syslog configuration on pfSense and confirmed that logs were actually being received on the Wazuh server's Syslog listener.
Custom Wazuh rule caused the manager to fail.	Corrected the XML syntax and replaced the invalid <code><field name="protocol"></code> condition with <code><match>icmp</match></code> .
Wazuh Manager failed to start after editing rules.	Validated the configuration using <code>wazuh-analysisd -t</code> , fixed the rule, and restarted the manager successfully.
Malware detection verification.	Used the EICAR test file to safely validate SquidClamAV functionality without introducing real malware into the lab.

Table 10: Challenges encountered during the project and how each was resolved.

11 Conclusion

This project successfully implemented an integrated security monitoring environment using pfSense Firewall, Squid Proxy, SquidClamAV, and Wazuh SIEM. Firewall events were centralized through Syslog, malware downloads were detected and blocked using the EICAR test file, and custom Wazuh rules generated alerts for monitored network activity. Beyond the individual technical steps, the exercise reinforced how the different layers of a SOC toolchain depend on one another: without the firewall generating logs, Wazuh would have nothing to analyze; without the custom rule, those logs would never surface as an actionable alert; and without SquidClamAV, malicious downloads would reach the client undetected. The completed implementation demonstrated end-to-end log collection, threat detection, and alert monitoring in a controlled virtual lab environment, and provided practical, hands-on experience with the tools and workflows used in a real-world Security Operations Center.

12 References

Only official, authoritative sources were used while researching and configuring each component of this lab:

1. Wazuh Documentation – <https://documentation.wazuh.com/>
2. pfSense Documentation – <https://docs.netgate.com/pfsense/>
3. Squid Proxy Documentation – <https://www.squid-cache.org/>
4. ClamAV Documentation – <https://docs.clamav.net/>
5. EICAR Anti-Malware Test File – <https://www.eicar.org/>

13 Appendix – Important Commands

The commands below were the most frequently used during configuration, debugging, and validation of the Wazuh Manager and its custom rule.

Purpose	Command
Restart Wazuh Manager	<code>sudo systemctl restart wazuh-manager</code>
Test Custom Rule	<code>sudo /var/ossec/bin/wazuh-logtest</code>
Validate Rules	<code>sudo /var/ossec/bin/wazuh-analysisd -t</code>
View Alerts	<code>sudo tail -f /var/ossec/logs/alerts/alerts.log</code>
Check Manager Status	<code>sudo systemctl status wazuh-manager</code>
Test Network	<code>ping 8.8.8.8</code>

Table 11: Frequently used commands for managing and validating the Wazuh Manager.